INCH-POUND

MIL-C-41833E(GL)
30 March 1990
SUPERSEDING
MIL-C-41833D
8 February 1985

MILITARY SPECIFICATION

COVERALL, MECHANIC'S, COLD WEATHER

This specification is approved for use by the Natick Research, Development, and Engineering Center, Department of the Army, and is available for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 Scope. This specification covers a coverall made from polyester and rayon gabardine cloth.

1.2 Classification. The coveralls shall be one type in the following sizes, as specified (see 6.2).

Schedule of sizes

Small
Medium
Large

2. APPLICABLE DOCUMENTS

2.1 Government documents.

* 2.1.1 Specifications, standards, and handbooks. The following specifications, standards, and handbooks form a part of this document to the extent specified herein. Unless otherwise specified, the issues of these documents are those listed in the issue of the Department of Defense Index of Specifications and Standards (DODISS) and supplement thereto, cited in the solicitation (see 6.2).

Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be of use in improving this document should be addressed to: U.S. Army Natick Research, Development, and Engineering Center, Natick, MA 01760-5014 by using the Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document or by letter.

AMSC N/A FSC 8415

DISTRIBUTION STATEMENT A. Approved for public release; distribution is unlimited.

Source: https://assist.dla.mil -- Downloaded: 2026-10-21T20:07Z
Check the source to verify that this is the current version before use.

SPECIFICATIONS

FEDERAL

V-F-106	-	Fasteners, Slide, Interlocking
A-A-203	-	Paper, Kraft, Untreated
V-T-285	-	Thread, Polyester
JJ-W-155	-	Webbing, Textile, (Cotton, Elastic)
DDD-L-20	-	Label: for Clothing, Equipage and Tentage (General Use)
PPP-B-636	-	Boxes, Shipping, Fiberboard

MILITARY

MIL-B-371	-	Braid, Textile, Tubular
MIL-C-508	-	Cloth, Oxford, Nylon, 3 Ounce
MIL-F-10884	-	Fasteners, Snap
MIL-L-35078	-	Loads, Unit: Preparation of Semiperishable Subsistence Items; Clothing, Personal Equipment and Equipage; General Specification For
MIL-C-41820	-	Cloth, Gabardine, Polyester and Rayon
MIL-C-43303	-	Cord, Elastic, Cotton
MIL-T-43548	-	Thread, Polyester Core: Cotton-, Rayon-, or Polyester-Covered

STANDARDS

FEDERAL

FED-STD-751	-	Stitches, Seams and Stitchings

MILITARY

MIL-STD-105	-	Sampling Procedures and Tables for Inspection by Attributes
MIL-STD-129	-	Marking for Shipment and Storage
MIL-STD-147	-	Palletized Unit Loads
MIL-STD-731	-	Quality of Wood Members for Containers and Pallets
MIL-STD-1668	-	Provisions for Evaluating Quality of Cloth Coveralls

(Unless otherwise indicated, copies of federal and military specifications, standards, and handbooks are available from the Standardization Documents Order Desk, Building 4D, 700 Robbins Avenue, Philadelphia, PA 19111-5094.)

Source: https://assist.dla.mil -- Downloaded: 2026-10-21T20:07Z
Check the source to verify that this is the current version before use.

* 2.2 Non-Government publications. The following documents form a part of this document to the extent specified herein. Unless otherwise specified, the issues of the documents which are DoD adopted are those listed in the issue of the DODISS cited in the solicitation. Unless otherwise specified, the issues of documents not listed in the DODISS are the issues of the documents cited in the solicitation (see 6.2).

AMERICAN SOCIETY FOR TESTING AND MATERIALS (ASTM)

D 3951 - Standard Practice for Commercial Packaging

(Application for copies should be addressed to the American Society for Testing and Materials, 1916 Race Street, Philadelphia, PA 19103-1187.)

THE COLOR ASSOCIATION OF THE UNITED STATES

Standard Color Card of America

Department of Defense Standard Color Card for Sewing Threads

(Color cards may be available from the Color Association of the United States, 343 Lexington Avenue, New York, NY 10016-0927. If color cards are not available from the Color Association, individual color samples may be obtained from the contracting activity or as directed by the contracting activity.)

(Non-Government standards and other publications are normally available from the organizations that prepare or distribute the documents. These documents also may be available in or through libraries or other informational services.)

2.3 Order of precedence. In the event of a conflict between the text of this document and the references cited herein, the text of this document takes precedence. Nothing in this document, however, supersedes applicable laws and regulations unless a specific exemption has been obtained.

3. REQUIREMENTS

* 3.1 First article. When specified (see 6.2), a sample shall be subjected to first article inspection (see 6.3) in accordance with 4.3.

3.2 Guide samples. Guide samples, when furnished, are solely for guidance and information to the contractor (see 6.4). Variation from this specification may appear in the sample, in which case this specification shall govern.

3.3 Material. It is encouraged that recycled material be used when practical as long as it meets the requirements of this specification.

Source: https://assist.dla.mil -- Downloaded: 2026-10-21T20:07Z
Check the source to verify that this is the current version before use.

3.3.1 Basic material. The basic material shall be polyester and rayon gabardine, Olive Green 106, conforming to type III, class 2 of MIL-C-41820.

3.3.2 Lining. The material for the lining shall be nylon oxford cloth, 3-ounce, Olive Green 106, conforming to type I, class 2, of MIL-C-508.

3.3.3 Elastic cord. The elastic cord for sleeves and leg bottoms shall be 3/16-inch diameter, natural, or Olive Drab 7, conforming to class 2 of MIL-C-43303.

3.3.4 Webbing, elastic. The elastic webbing for back waist shall be 1-1/2 inches wide, unbleached or Olive Drab 7, conforming to type I, class 10 of JJ-W-155 except that the requirement for colorfastness to accelerated weathering shall not apply.

* 3.3.5 Slide fasteners. The slide fasteners shall have either a brass or extruded individual element plastic chain in accordance with V-F-106. The slide fastener tapes and thongs of both chain types shall be Olive Drab, Shade S, Cable No. 66519 and the chain, slider, and stops of the extruded plastic chain type shall approximately match the tape shade. The tapes and thongs shall show "good" fastness to laundering. Thongs are required on all slide fasteners and their finished lengths shall be measured from the top of the loop to the end of the thong. All thong tape ends shall either be heat-cut or bartack stitched to prevent fraying.

* 3.3.5.1 Front opening. The slide fastener for the front opening shall conform to type I, style 3 with the brass chain as size MHS and the extruded plastic chain as size H in accordance with V-F-106. The slider pull shall have a hole large enough to accommodate the required thong, and the finished thong length shall be 3 to 3-1/2 inches long. The length of slide fastener for all end item sizes shall be 34-1/2 inches long.

* 3.3.5.2 Hip pocket. The slide fastener for the hip pocket shall conform to type I, style 3, with the brass chain as size MS and the extruded plastic chain as size MH in accordance with V-F-106. The slider shall be equipped with a long tab pull and required thong with a finished length of 2 to 2-1/2 inches long. The length of slide fastener for all end item sizes shall be 8-1/2 inches long.

3.3.6 Snap fasteners. The snap fasteners for the front flap shall consist of a size 1 or 2 button with socket for the female half and a size 1 or 2 eyelet with stud for the male half conforming to style 2A, finish 2 of MIL-F-10884.

3.3.7 Thread. The thread for seaming and stitching shall be polyester size B and E, as indicated in table II, shade S-1, C.A. No. 66022, conforming to type I, class 1 of V-T-285. Cotton-covered polyester thread conforming to MIL-T-43548 may be substituted for the polyester thread. When this thread is

Source: https://assist.dla.mil -- Downloaded: 2026-10-21T20:07Z
Check the source to verify that this is the current version before use.

used, the substitution shall be 30, 3 ply for the needle thread of the 401 stitch type; 50, 3 ply for the 301 stitch type and for the bobbin thread of the 401 stitch type; and 70, 2 ply for overedging.

3.3.7.1 Colorfastness. The dyed thread shall show fastness to light and laundering equal to or better than the standard sample. When no standard sample is available, the dyed thread shall show "good" fastness to light and laundering.

* 3.3.8 Labels. Each coverall shall have a class 1 identification label, a class 2 size label, and a class 3 instruction label conforming to type VI of DDD-L-20. As alternatives, a class 3 combination identification and instruction label or a class 14 combination size, identification, and instruction label may be used. The labels shall show fastness to drycleaning. The instruction label or combination label, as applicable, shall include the following information:

1. Coverall may be worn over intermediate layers of the cold-dry combat uniform.
2. Launder in accordance with laundry formula E in fixed launderies or laundry formula I in field launderies.

DO NOT STARCH

DO NOT REMOVE THIS LABEL

3.3.9 Braid-hanger loop. The braid for the hanger loop shall be either cotton or rayon, OG 106, conforming to type IX, class 1 of MIL-B-371 (see operation 1.d. for alternate).

3.4 Design. The coverall has a bi-swing back with elastic at the waist, a right hip pocket, side hand entries, and a front opening from the neck to the middle of the thigh. The hip pocket has a slide fastener with inside and outside protective welts. The front opening has a slide fastener and a protective flap with snap fasteners. The bottom of the sleeves and legs have elastic in the hems. The right and left front of the trousers are double ply and the jacket part is lined.

3.4.1 Figure. Figure 1 is furnished for information only. To the extent of any inconsistencies between the written specification and the figure, the specification shall control.

3.5 Patterns. Standard patterns to be used to cut working patterns will be furnished by the Government. The patterns provide a 1/2-inch seam allowance for side seams, inseams, seat seam, and waist seam of the coverall and the lining and 3/8 inch for all other seams. The working patterns shall be identical to the Government patterns, which shall not be altered in any way.

Source: https://assist.dla.mil -- Downloaded: 2026-10-21T20:07Z
Check the source to verify that this is the current version before use.

3.5.1 Pattern parts. The component parts of the coverall shall be cut from the materials as specified in accordance with the number of parts required as shown in table I.

TABLE I. List of pattern parts

Material	Nomenclature of parts	Cut parts
	Jacket parts	
Basic material (see 3.3.1)	Back	1
	Right front	1
	Left front	1
	Right front facing	1
	Left front facing	1
	Front sleeve (with side body)	2
	Back sleeve (with side body)	2
	Collar	2
	Front flap	1
	Trouser parts	
Basic material (see 3.3.1)	Right front (outer layer)	1
	Right front (inner layer)	1
	Left front (outer layer)	1
	Left front (inner layer)	1
	Left fly panel	2
	Back	2
	Side opening bearer	2
	Hip pocket	1
	Hip pocket welt	2
	Hip pocket bearer	1
	Lining parts	
Nylon oxford cloth (see 3.3.2)	Right front	1
	Left front	1
	Back	1
	Front sleeve (with side body)	2
	Back sleeve (with side body)	2
	Back waist elastic cover	1

Source: https://assist.dla.mil -- Downloaded: 2026-10-21T20:07Z
Check the source to verify that this is the current version before use.

3.6 Construction.

3.6.1 Stitches, seams and stitchings. Stitch, seam and stitching types specified in table II shall conform to FED-STD-751. When two or more methods of seaming or type of stitch are given for the same operation, any one of them may be used. Where stitch type 401 is used, the looper (underthread) shall be on the inside of the coverall. The gage of stitching for double-lapped, double-stitched seams shall be 1/4 to 5/16 inch. The gage of all overedge stitching shall be 3/16 inch and the seaming row, when using safety stitching, shall be 3/16 inch from back edge of overedge stitch. All seams shall start and finish evenly. Seams required to be worked out, that have a fold of 1/16 inch or more in depth, shall be considered a defect.

3.6.1.2 Seam type. Seam types shall be as specified in table II with seam allowances maintained and seams properly sewn so that no raw edges, runoffs, twists, pleats, puckers, or open seams shall result.

3.6.1.3 Type 301 stitching. Ends of all stitching shall be backstitched or overstitched 1/4 inch minimum, except when caught in other stitching or turned under in a hem. Thread tension shall be maintained so that there will be no loose stitching resulting in loose bobbin or top thread, or excessively tight stitching resulting in puckering of the materials sewn. The lock shall be embedded in the materials sewn.

3.6.1.3.1 Repairs of type 301 stitching. Repairs of type 301 stitching shall be as follows:

a. When thread breaks or bobbin run-outs occur during sewing, the stitching shall be repaired by restarting the stitching a minimum of 1/2 inch back of the end of the stitching. 1/

b. Thread breaks, two or more consecutive skipped or run-off stitches noted during inspection of the item (in-process or end item) shall be repaired by overstitching. The stitching shall start a minimum of 1/2 inch in back of the defective area, continue over the defective area, and continue a minimum of 1/2 inch beyond the defective area on to the existing stitching. Loose or excessively tight stitching shall be repaired by removing the defective stitching, without damaging the materials and restitching in the required manner. 1/

1/ When making the above repairs, the ends of the stitching are not required to be backstitched.

3.6.1.4 Type 401, 502, 503, 504, 515, 516, and 519 stitching. Thread tension shall be maintained so that there will be no loose stitching. All repairs shall be in accordance with 3.6.1.3.1a. and 3.6.1.3.1b. Thread tension shall be maintained so that there will be no loose or excessively tight stitching resulting in puckering of the materials sewn.

Source: https://assist.dla.mil -- Downloaded: 2026-10-21T20:07Z
Check the source to verify that this is the current version before use.

3.6.2 Bartacking. Unless otherwise specified, bartacks shall be as follows:

		Tolerance		
Length (inches)	Width (inches)	Length (inches)	Width (inches)	Stitches per bartack
1/2	1/8	± 1/16	± 1/32	28
3/4	1/8	± 1/16	± 1/32	42
1	1/8	± 1/16	± 1/32	42

Bartacking shall be free from thread breaks and loose stitching.

3.6.3 Thread ends. All thread ends shall be trimmed to 1/4 inch maximum length.

3.7 Manufacturing operations requirements. The coverall shall be manufactured in accordance with operation requirements specified in table II. The contractor is not required to follow the exact sequence of operations. Any additional basting or holding stitching used to facilitate manufacture is permissible provided that the thread is removed or does not show on the finished coverall.

Source: https://assist.dla.mil -- Downloaded: 2026-10-21T20:07Z
Check the source to verify that this is the current version before use.

NO.	TABLE II. MANUFACTURING OPERATIONS REQUIREMENTS	STITCH TYPE	SEAM AND STITCHING TYPE	STITCHES PER INCH	THREAD		
					NEEDLE	BOBBIN/ LOOPER	COVER
1.	Cutting. a. Cut the coverall in accordance with patterns furnished, which show directional lines, marks for placement of pocket, and notches for the proper assembly of parts. b. Cut all basic material parts from the same piece of material, except front facings, undercollar, inner layer of trouser fronts, side opening bearers, hip pocket, hip pocket welts, hip pocket bearer, and hanger loop (see 3.3.9 for alternate braid and operation 1.d. for finished width), which may be cut from edges. Parts cut from ends shall approximate the shade of the coverall. c. Place fronts and backs (inner and outer layer of trouser parts only) so that the directional line is in the warp direction. Directional lines on the fronts and backs may vary from the warp direction by not more than 1-1/2 inches. Measurements shall be taken at the top and bottom edges of the pattern from the directional lines to the selvaged edge (on the even side of lay) of the fabric, and the difference between the two measurements shall not exceed the tolerance specified. d. Cut stripping for hanger loop from basic material or lining material of sufficient width to finish 1/4 to 1/2 inch wide (see operation 6).						

NATICK Form 903
1 Dec 76 EDITION OF 1 OCT 76 WILL BE USED UNTIL EXHAUSTED.

Source: https://assist.dla.mil - Downloaded: 2026-10-21T20:07Z
Check the source to verify that this is the current version before use.

NO.	TABLE II. (cont'd) MANUFACTURING OPERATIONS REQUIREMENTS	STITCH TYPE	SEAM AND STITCHING TYPE	STITCHES PER INCH	THREAD		
					NEEDLE	BOBBIN/ LOOPER	COVER
1.	Cutting. (cont'd) e. Cut lining parts from same piece of material, except waist elastic cover, which may be cut from ends. f. Cut the 3/16-inch elastic cord for the bottom of sleeves and legs of sufficient length so that in the finished garment (see operation 18 and 20) they will measure as follows: Sleeves - 4-1/2 inches ± 1/2 inch (from folded edge to folded edge). Legs - 7 inches ± 1/2 inch (from folded edge to folded edge). g. Cut the 1-1/2 inch elastic webbing for the back waist of sufficient length to measure as follows in the finished garment (see operation 16 and table III). The measurement shall be taken from one end of the elastic to the other end. Size, small - 15-1/2 inches ± 1/2 inch Size, medium - 17-1/2 inches ± 1/2 inch Size, large - 19-1/2 inches ± 1/2 inch						
2.	Replacement of defective or damaged parts. Care shall be exercised during the spreading, cutting and manufacturing operations to assure that material defects or damages, as classified in MIL-STD-1668, are excluded and replaced with non-defective and properly matched material.						

NATICK Form 903
1 Dec 76 EDITION OF 1 OCT 76 WILL BE USED UNTIL EXHAUSTED.

Source: https://assist.dla.mil - Downloaded: 2026-10-21T20:07Z
Check the source to verify that this is the current version before use.

NO.	TABLE II. (cont'd) MANUFACTURING OPERATIONS REQUIREMENTS	STITCH TYPE	SEAM AND STITCHING TYPE	STITCHES PER INCH	THREAD		
					NEEDLE	BOBBIN/ LOOPER	COVER
3.	Marking. a. Mark, ticket, or bundle all component parts of the coverall except those parts cut from ends to ensure correct size, proper assembly, and a uniform shade throughout the coverall. b. Any method of marking may be used except: (1) Metal fastening devices. (2) Sew-on tickets. (3) Adhesive type tickets that leave traces of adhesive on the material after removal of the tickets. c. The use of ink pad numbering machine, rubber stamp, or pencil will be acceptable provided the numbers do not show through to the outside of the coverall and the numbers are covered by the seam allowance.						
4.	Attach identification and instruction label or combination label. Position the identification and instruction label or combination label on the hip pocket, with the label facing the wearer when the pocket is in the finished position. The top of the instruction label may be caught in the stitching at the bottom of the identification label. Stitch on all four sides. The stitching shall not be through the printing. Stitching of subsequent operations shall not be through the label.	301 or 401 or 101	LSbj-1 LSbj-1 LSbj-1	8-14 8-14 8-14	B E B	B B	

NATICK Form 903
1 Dec 76 EDITION OF 1 OCT 76 WILL BE USED UNTIL EXHAUSTED.

Source: https://assist.dla.mil - Downloaded: 2026-10-21T20:07Z
Check the source to verify that this is the current version before use.

TABLE II. (cont'd)

NO.	MANUFACTURING OPERATIONS REQUIREMENTS	STITCH TYPE	SEAM AND STITCHING TYPE	STITCHES PER INCH	THREAD		
					NEEDLE	BOBBIN/ LOOPER	COVER
5.	Making lining and attach size label.						
	a. Fold the center back pleat (box or inverted box) as indicated by marks on pattern; center size label over pleat with the top of the label not more than 1 inch from the neck raw edge. Stitch on all four sides. The label may be positioned so that the top of the label will be caught in the stitching when stitching down the top collar (operation 17). The stitching shall not be through the printing.	301 or 401 or 101	LSbj-1 LSbj-1 LSbj-1	8-14 8-14 8-14	B E B	B B	
	b. Stitch front sleeve to back sleeve at back arm.	301 or 401 515 or 516 or 519	LSc-2 LSc-2 or SSa-2	10-14 10-14 10-14	B E E (chain stitching) B (over-edging stitching)	B B B B	

NATICK Form 903
1 Dec 76 EDITION OF 1 OCT 76 WILL BE USED UNTIL EXHAUSTED.

Source: https://assist.dla.mil - Downloaded: 2026-10-21T20:07Z
Check the source to verify that this is the current version before use.

NO.	TABLE II. (cont'd) MANUFACTURING OPERATIONS REQUIREMENTS	STITCH TYPE	SEAM AND STITCHING TYPE	STITCHES PER INCH	THREAD		
					NEEDLE	BOBBIN/ LOOPER	COVER
5.	Make lining and attach size label. (cont'd)						
	c. Join facings (basic material) to the lining fronts, join shoulders, and stitch sleeves to body. Raise stitch seams 3/16 to 1/4 inch from joining seams with the facing-to-front joining seams turned toward sleeves, shoulder seams toward back, and sleeves-to-body seams turned toward body.	515 or 516 or	LSbm-3(a)	8-14	E (chain stitch-ing)	B	
		519			B (over-edging stitch-ing)	B	
			and				
		301 or	LSbm-3(b)	10-14	B	B	
		401	LSbm-3(b)	10-14	E	B	
	d. Stitch front sleeve to back sleeve at underarm.	301 or	LSc-2	10-14	B	B	
		401	LSc-2	10-14	E	B	
			or				
		515 or 516	SSa-2	10-14	E (chain stitch-ing)	B	
		519			B (over edging stitch-ing)	B	
6.	Make hanger loop.						
	a. Fold stripping with the raw edges turned in, and stitch 1/16 to 1/8 inch from each edge.	301	EFp-2	8-14	B	B	
	or						

NATICK Form 903
1 Dec 76 EDITION OF 1 OCT 76 WILL BE USED UNTIL EXHAUSTED.

Source: https://assist.dla.mil - Downloaded: 2026-10-21T20:07Z
Check the source to verify that this is the current version before use.

TABLE II. (cont'd)

NO.	MANUFACTURING OPERATIONS REQUIREMENTS	STITCH TYPE	SEAM AND STITCHING TYPE	STITCHES PER INCH	THREAD		
					NEEDLE	BOBBIN/ LOOPER	COVER
6.	Make hanger loop. (cont'd)						
	b. Turn in the edges of two pieces of stripping, and stitch together with the stitching 1/16 to 1/8 inch from each edge.	301	EFn-2	8-14	B	B	
	NOTE: A selvage edge need not be turned in.						
7.	Make collar.						
	Stitch top collar and under collar together, trim if necessary; force out points and edges, turn, and edge stitch 3/16 to 1/4 inch from edge leaving each end unstitched 1 to 1-1/2 inches to facilitate setting collar.	301	SSe-2(a)	8-14	B	B	
		401 and	SSe-2(a)	8-14	E	B	
		301	SSe-2(b)	10-14	B	B	
8.	Make front protective flap.						
	a. Fold flap in half lengthwise; stitch across both ends; trim corners, if necessary; force out edges and corners, and edge stitch each end and the folded edge 3/16 to 1/4 inch from edge.	301 or	SSe-2(a)	8-14	B	B	
		401 and	SSe-2(a)	8-14	E	B	
		301	SSe-2(b) and	10-14	B	B	
		301	OSf-1	10-14	B	B	
	b. Overedge the back raw edges together.	502 or	SSa-1	8-14	B	B	
		503 or	SSa-1	8-14	B	B	
		504	SSa-1	8-14	B	B	
	c. Stitch the left half of the slide fastener to the back edge of the flap 1/8 to 1/4 inch from edge and with the top of the tape turned back to be caught in a subsequent operation. The top of the chain shall be not more than 5/8 inch from the top of flap.	301	SSa-1	10-14	B	B	
		401	SSa-1	10-14	E	B	

Source: https://assist.dla.mil - Downloaded: 2026-10-21T20:07Z
Check the source to verify that this is the current version before use.

TABLE II. (cont'd)

NO.	MANUFACTURING OPERATIONS REQUIREMENTS	STITCH TYPE	SEAM AND STITCHING TYPE	STITCHES PER INCH	THREAD		
					NEEDLE	BOBBIN/ LOOPER	COVER
8.	Make front protective flap. (cont'd)						
	d. Attach the four socket parts of the snap fasteners to the flap through both plies positioned as indicated on pattern (± 1/8 inch off center tolerance).						
9.	Make bi-swing back (outer shell).						
	a. Fold each side of back along the outer foldline indicated on pattern with material face-to-face, and stitch 1/16 to 1/4 inch from folded edge.	301	OSf-1	8-14	B	B	
	b. Fold each side of back along inner foldline indicated on pattern with the material back to back, and stitch 1/16 to 1/4 inch from folded edge.	301 401	OSf-1 OSf-1	8-14 8-14	B E	B B	
10.	Assemble jacket (outer shell).						
	a. Stitch front sleeves to back sleeves at back arm; turn seams toward front sleeves, and raise stitch 3/16 to 1/4 inch from joining seams or stitch front sleeves to back sleeves at back arm with a double lapped and double stitched seam.	301 or 401 or 401	LSq-2 LSq-2 LSc-2	10-14 10-14 10-14	B E E	B B B	
	b. Join shoulders, catching the bi-swing pleats in the stitching; turn seams toward back, and raise stitch 3/16 to 1/4 inch from joining seams or join shoulders, catching the bi-swing pleats in the stitching with a double-lapped and double-stitched seam.	301 401 or 401	LSq-2 LSq-2 LSc-2	10-14 10-14 10-14	B E E	B B B	

NATICK Form 903
1 Dec 76 EDITION OF 1 OCT 76 WILL BE USED UNTIL EXHAUSTED.

Source: https://assist.dla.mil - Downloaded: 2026-10-21T20:07Z
Check the source to verify that this is the current version before use.

NO.	TABLE II. (cont'd) MANUFACTURING OPERATIONS REQUIREMENTS	STITCH TYPE	SEAM AND STITCHING TYPE	STITCHES PER INCH	THREAD		
					NEEDLE	BOBBIN/ LOOPER	COVER
10.	Assemble jacket (outer shell). (cont'd)						
	c. Stitch sleeves to body; turn seams toward body, and raise stitch 3/16 to 1/4 inch from joining seams.	301 or 401	LSq-2 LSq-2	10-14 10-14	B E	B B	
	d. Stitch front sleeves to back sleeves at underarm; turn seams toward front sleeves, and raise stitch 3/16 to 1/4 inch from joining seam or stitch front sleeve to back sleeve with a double-lapped and double-stitched seam.	301 or 401 or 401	LSq-2 LSq-2 LSc-2	10-14 10-14 10-14	B E E	B B B	
11.	Make hip pocket.						
	a. Overedge bottom of bearer if not selvage. (See operation 11d.)	502 or 503 or 504	EFd-1 EFd-1 EFd-1	8-14 8-14 8-14	B B B	B B B	
	b. Position top of bearer to top of pocketing as indicated by pattern, and stitch 1/8 to 1/4 inch from top of bearer.	301 or 401 or 101	LSbj-1 LSbj-1 LSbj-1	3-14 3-14 3-14	B E B	B B	
	c. If bottom of bearer is overedged or selvage, stitch to pocketing 1/16 to 3/16 inch from edge.	301 or 401	LSbj-1 LSbj-1	8-14 8-14	B E	B B	
	or						
	d. If bottom of bearer is not overedged or selvage, turn in 1/4 to 3/8 inch and stitch to pocketing 1/16 to 3/16 inch from turned-in edge.	301 or 401	LSd-1 LSd-1	8-14 8-14	B E	B B	
	e. Fold sides (length) of welts as indicated by pattern, and stitch 1/16 to 1/8 inch from raw edges. This operation may be omitted if sides are precreased.	301 or 401 or 101	EFa-1 EFa-1 EFa-1	3-14 3-14 3-14	B E B	B B	

Source: https://assist.dla.mil - Downloaded: 2026-10-21T20:07Z
Check the source to verify that this is the current version before use.

TABLE II. (cont'd)

NO.	MANUFACTURING OPERATIONS REQUIREMENTS	STITCH TYPE	SEAM AND STITCHING TYPE	STITCHES PER INCH	THREAD		
					NEEDLE	BOBBIN/ LOOPER	COVER
11.	Make hip pocket.						
	f. Fold welts in half lengthwise; insert each half of the slide fastener between a welt and with the welts extending 1/8 to 3/16 inch beyond the edge of the opened chain; stitch the length of each folded welt at center catching the slide fastener tapes in the stitching. When slide fastener is closed, the welts shall overlap 1/8 to 3/16 inch, and on the finished pocket the top welt shall overlap the lower one.	301 or 401 or 101	SSv-1 SSv-1 SSv-1	8-14 8-14 8-14	B E B	B B	
	g. Stitch bottom edge of pocketing to bottom of welt using 1/4 inch seam allowance, and with the seam turned toward pocketing raise stitch 3/16 to 1/4 inch from joining seam. When the pocket is finished, the seam allowances shall be between the pocket and the coverall.	301 or 401 and 301	SSe-2(a) SSe-2(a) SSe-2(b)	10-14 10-14 10-14	B E B	B B B	
	h. Cut pocket opening as indicated by pattern, and tongue notch edges 3/8 to 1/2 inch to allow a 3/4 to 1 inch wide finished opening.						
	i. Position the slide fastener (with welts) under the pocket opening with the top of the chain facing the seat seam. Turn in the bottom edge of the opening, and stitch through all plies, 1/16 to 1/8 inch from turned-in edge.	301	LSd-1	10-14	B	B	

Source: https://assist.dla.mil - Downloaded: 2026-10-21T20:07Z
Check the source to verify that this is the current version before use.

NO.	TABLE II. (cont'd) MANUFACTURING OPERATIONS REQUIREMENTS	STITCH TYPE	SEAM AND STITCHING TYPE	STITCHES PER INCH	THREAD		
					NEEDLE	BOBBIN/ LOOPER	COVER
11.	Make hip pocket. (cont'd)						
	j. Fold pocketing to its finished position, and safety stitch sides together.	515 or 516 or 519	SSa-2	10-14	E (chain stitching) and B (over edging stitching)	B B	
	k. With the pocket smooth and flat, turn in the top raw edge and the tongues at ends, and stitch through all plies 1/16 to 1/8 inch from edge, closing top of pocket.	301	LSd-1	10-14	B	B	
	l. Place two 1/2 inch vertical bartacks or a single 1 inch bartack at each side of hip pocket opening through fabric and ends of slide fastener tapes. The bartacks shall extend the full width of the pocket opening.	Bartack	--	28 per bartack or 42 per bartack	B B	B B	
12.	Attach hand-opening bearers to trouser backs.						
	a. Fold hand-opening bearers in half lengthwise and with the side seam allowances turned in over-edge across bottom.	502 or 503 or 504	SSa-1 SSa-1 SSa-1	8-14 8-14 8-14	B B B	B B B	

Source: https://assist.dla.mil -- Downloaded: 2026-10-21T20:07Z
Check the source to verify that this is the current version before use.

TABLE II. (cont'd)

NO.	MANUFACTURING OPERATIONS REQUIREMENTS	STITCH TYPE	SEAM AND STITCHING TYPE	STITCHES PER INCH	THREAD		
					NEEDLE	BOBBIN LOOPER	COVER
12.	Attach hand-opening bearers to trouser backs. (cont'd)						
	b. Position a bearer on each trouser back as indicated by pattern, and stitch to trouser from top to notch.	301 or 401	SSa-1 SSa-1	8-14 8-14	B E	B B	
	c. Turn bearer to its finished position, turn in the raw edge, and stitch bearer to trouser through all plies, 1/16 to 1/8 inch from turned-in edge.	301 or 401		10-14 10-14	B E	B B	
13.	Make trouser fronts.						
	a. To form top part of hand openings, superimpose the inner and outer layers of the trouser fronts, and stitch together from waist to side opening notch. Slash seam allowance at notch, turn to finished position, work out edges, and edge stitch 3/16 to 1/4 inch from edge. On the finished coverall, the face of the material of the inner layer may face the wearer or the outer layer.	301 or 401 and 301	SSe-2(a) SSe-2(b) SSe-2(b)	8-14 8-14 10-14	B E B	B B B	
	b. With edges even, overedge the inner and outer layer of the left trouser front together from waist to bottom of the front opening, and overedge the inner and outer ply of the front opening of the fly panel together.	502 or 503 or 504	SSa-1 SSa-1 SSa-1	8-14 8-14 8-14	B B B	B B B	
	c. With the edges even, baste or stitch the inseam and outseam seam allowances of the inner and outer layer of trouser fronts together and the seam allowances of the two sides of the fly panel. These operations may be omitted if there are no twists of fullness between the plies in the finished coverall.	301 or 401 or 101	SSa-1 SSa-1 SSa-1	2-14 2-14 2-14	B E B	B B	

NATICK Form 903
1 Dec 76 EDITION OF 1 OCT 76 WILL BE USED UNTIL EXHAUSTED.

Source: https://assist.dla.mil - Downloaded: 2026-10-21T20:07Z
Check the source to verify that this is the current version before use.

TABLE II. (cont'd)

NO.	MANUFACTURING OPERATIONS REQUIREMENTS	STITCH TYPE	SEAM AND STITCHING TYPE	STITCHES PER INCH	THREAD		
					NEEDLE	BOBBIN/ LOOPER	COVER
14.	Assemble trousers.						
	a. Join seat seam, front seam (crotch), and outseams with safety stitching. Turn seat seam toward left back (as worn), front seam toward right trouser, and outseams toward front, and raise stitch 3/16 to 1/4 inch from each joining seam or join seat seam and outseams with a double lapped and double stitched seam. In the seat seam, the right back shall overlap the left back; in the outseams, the two front panels shall overlap the two back panels.	515 or	LSbm-3(a)	8-14	E (chain stitching)	B	
		516 or 519			B (over edging stitching)	B	
		and 301 or	LSbm-3(b)	10-14	B	B	
		401 or	LSbm-3(b)	10-14	E	B	
		401	LSc-2	10-14	E	B	
	b. Stitch across bottom of side hand-openings through front of coverall and bearer for distance of 2 inches ± 1/2 inch, securely tacking ends of stitching.	301	SSv-1	10-14	B	B	
15.	Attach jacket to trousers.						
	a. Stitch jacket to trouser, forming a pleat on front as indicated by pattern. The outside folded edge of the pleats shall face the sides of the jacket.	301 or	SSa-1	10-14	B	B	
		401	SSa-1	10-14	B	B	
	b. When alternate method of closing bottom of lining is used (see operation 16.f.), turn jacket up to finished position, and raise stitch 1/16 to 1/8 inch from turned edge.	301	LSq-2(b)	10-14	B	B	

NATICK Form 903
1 Dec 76 EDITION OF 1 OCT 76 WILL BE USED UNTIL EXHAUSTED.

Source: https://assist.dla.mil -- Downloaded: 2026-10-21T20:07Z
Check the source to verify that this is the current version before use.

TABLE II. (cont'd)

NO.	MANUFACTURING OPERATIONS REQUIREMENTS	STITCH TYPE	SEAM AND STITCHING TYPE	STITCHES PER INCH	THREAD		
					NEEDLE	BOBBIN/ LOOPER	COVER
16.	Attach lining and protective flap.						
	a. Position the left (as worn) part of the lining to the jacket with the protective flap (with slide fastener attached) placed between the front facing (attached to lining) and jacket, and stitch across step from notch and down front. Turn seam allowance at bottom of facing and the jacket-to-trousers joining seam upward, and continue stitching the flap to the trouser down to end of opening. The top of the flap shall be not more than 1/4 inch from top of step.	301 or 401 301 401	SSq-2(a) SSq-2(a) and SSa-1 SSa-1	8-14 8-14 8-14 8-14	B E B E	B B B B	
	b. Position the right (as worn) facing part of the lining to the jacket, and stitch across step from notch and down front with the seam allowance of lining (facing) and jacket-to-trousers joining seams turned upward.	301 401	SSe-2(a) SSe-2(a)	8-14 8-14	B E	B B	
	c. Slash seam allowance at step notch; turn lining to its finished position, force out edges, and raise stitch the jacket-to-facing and flap-to-trouser joining seams using 1/8 to 1/4 inch gage.	301 301	SSe-2(b) and SSq-2(b)	10-14 10-14	B B	B B	
	d. Slash seam allowance at step notch; with lining in its finished position, force out edge, and edge stitch right front across step and down front using 1/8 to 1/4 inch gage, and with the seam allowance of the trousers turned in, continue stitching to bottom of opening.	301 301	SSe-2(b) and EFa-1	10-14 10-14	B B	B B	

NATICK Form 903
1 Dec 76 EDITION OF 1 OCT 76 WILL BE USED UNTIL EXHAUSTED.

Source: https://assist.dla.mil - Downloaded: 2026-10-21T20:07Z
Check the source to verify that this is the current version before use.

TABLE II. (cont'd)

NO.	MANUFACTURING OPERATIONS REQUIREMENTS	STITCH TYPE	SEAM AND STITCHING TYPE	STITCHES PER INCH	THREAD		
					NEEDLE	BOBBIN/ LOOPER	COVER
16.	Attach lining and protective flap. (cont'd)						
	e. Stitch bottom of lining to coverall at waist, from inside, forming front pleats on lining as indicated by pattern. The stitching shall superimpose or be adjacent to the trouser-to-jacket joining seam. The outside folded edge of the pleats shall face the fronts. With the seam turned toward the jacket, raise stitch from the outside of coverall, 3/16 to 1/4 inch from the jacket-to-trousers joining seam.	301 or 401 301	SSq-2(a) SSq-2(a) and SSq-2(b)	8-14 8-14 10-14	B E B	B B B	
	or						
	f. Turn under bottom hem of lining, forming front pleats on lining as indicated by pattern, and stitch to coverall 1/8 to 1/4 inch from turned edge of lining. The outside folded edge of the pleats shall face the front.	301	LSb-1	10-14	B	B	
	g. Stitch lining to neck of coverall, catching the hanger loop in the stitching. In the finished coverall, the distance between the inside edge of the hanger loop shall be 2-3/4 inches ± 1/4 inch, and the hanger loop shall be not more than 3/8 inch off center.	301	SSa-1	8-14	B	B	
	h. Position back elastic at waist, centered over the waist seam, and stitch to each side of back adjacent to the side seams using box stitching or two vertical bartacks on each side. The bartacks shall be 3/4 inch, placed one above the other, and 1/8 to 3/16 inch from end of elastic. The box patterns shall be centered on the elastic	301 or bartack		10-14 42 per bartack	B B	B B	

NATICK Form 903
1 Dec 76 EDITION OF 1 OCT 76 WILL BE USED UNTIL EXHAUSTED.

Source: https://assist.dla.mil Downloaded: 2026-10-21T20:07Z
Check the source to verify that this is the current version before use.

TABLE II. (cont'd)

NO.	MANUFACTURING OPERATIONS REQUIREMENTS	STITCH TYPE	SEAM AND STITCHING TYPE	STITCHES PER INCH	THREAD		
					NEEDLE	BOBBIN/ LOOPER	COVER
16.	Attach lining and protective flap. (cont'd)						
	and shall measure 1-3/16 to 1-1/4 inches in length by 11/16 to 3/4 inch in width. An automatic box stitching machine may be used.						
	i. Turn in all raw edges of the cover strip, position over the elastic webbing, and stitch to coverall 1/16 to 1/8 inch from each edge.	301	LSd-1	10-14	B	B	
17.	Set collar.						
	a. Stitch undercollar to neck, catching the ends of the hanger loop in the stitching.	301 or 401	SSa-1 SSa-1	8-14 8-10	B B	B B	
	b. Turn in raw edges of top collar, and stitch down through all plies with two rows of stitching 1/16 to 1/8 inch apart. Continue the edge stitching on collar (see operation 7).	301		10-14	B	B	
18.	Finish sleeves.						
	a. Fit lining into sleeves with seams of lining and jacket matching, and overedge linings to sleeves at wrist (see operation 18.c).	502 503 or 504	SSa-1 SSa-1 SSa-1	8-14 8-14 8-14	B B B	B B B	
	b. Overlap ends of elastics, and tack or 3/4 inch bartack to inseams of sleeves.	301 or bartack or 301		10-14 42 per bartack 42 per tack	B B B	B B B	

NATICK Form 903
1 Dec 76 EDITION OF 1 OCT 76 WILL BE USED UNTIL EXHAUSTED.

Source: https://assist.dla.mil -2B downloaded: 2026-10-21T20:07Z
Check the source to verify that this is the current version before use.

TABLE II. (cont'd)

NO.	MANUFACTURING OPERATIONS REQUIREMENTS	STITCH TYPE	SEAM AND STITCHING TYPE	STITCHES PER INCH	THREAD		
					NEEDLE	BOBBIN/ LOOPER	COVER
18.	Finish sleeves. (cont'd)						
	c. Turn up bottom of sleeve (and if not overedged in operation 18.a, turn in raw edge). Enclose elastic within the fold, and stitch 1/16 to 1/8 inch from turned edge of back edge of overedge stitching, whichever is applicable, forming hem. The hem shall finish 3/4 to 1 inch wide.	301	EFa-1	10-14	B	B	
			or				
		301	EFb-1	10-14	B	B	
19.	Attach right half of slide fastener.						
	Position the right half of slide fastener on right front with the edge of the scoops 3/8 to 1/2 inch from the front edge; turn in tape at top of opening, and stitch to coverall through all plies with two rows of stitching. One row shall be 1/2 to 5/8 inch from back edge of scoops, and one row 1/16 to 1/8 inch from back edge of tape.	301		10-14	B	B	
20.	Finish trousers.						
	a. Join bottom of fly panel to trousers with safety stitching, catching the end of the slide fastener tape in the stitching; turn seam towards bottom of trousers, and raise stitch 3/16 to 1/4 inch from joining seam.	515 or	LSbm-3(a)	8-14	E (chain stitching)	B	
		516 or 519 and			B (overedging stitching)	B	
		301 or	LSbm-3(b)	10-14	B	B	
		401	LSbm-3(b)	10-14	E	B	

NATICK Form 903
1 Dec 76 EDITION OF 1 OCT 76 WILL BE USED UNTIL EXHAUSTED.

Source: https://assist.dla.mil - Downloaded: 2026-10-21T20:07Z
Check the source to verify that this is the current version before use.

TABLE II. (cont'd)

NO.	MANUFACTURING OPERATIONS REQUIREMENTS	STITCH TYPE	SEAM AND STITCHING TYPE	STITCHES PER INCH	THREAD NEEDLE	THREAD BOBBIN LOOPER
20.	Finish trousers. (cont'd)					
	b. Join inseams in one operation with safety stitching. Turn seam toward back of trousers, and raise stitch 3/16 to 1/4 inch from joining seam or join inseam of trousers with a double-lapped, double-stitched seam with the fronts over the backs.	515 or	LSbm-3(a)	8-14	E (chain stitching)	B
		516 or 519 and			B (overedging stitching)	B
		301 or	LSbm-3(b)	10-14	B	B
		401 or	LSbm-3(b)	10-14	E	B
		401	LSc-2	10-14	E	B
	c. Overlap ends of elastics, and tack or 3/4 inch bartack to inseams of legs.	301 or bartack		10-14	B	B
				42 per bartack	B	B
		301		42 per tack	B	B
	d. Turn up bottom of legs, turn in raw edge and stitch 1/16 to 1/8 inch from edge, enclosing the elastic within the hem. The hem shall finish 1 inch to 1-1/4 inches wide.	301	EFb-1	10-14	B	B
21.	Attach stud part of snap fastener.					
	Attach the stud part of the snap fastener on right side of the coverall aligned with the socket part on flap. The flap shall lie smooth and flat when snap fasteners are engaged.					

NATICK Form 903
1 Dec 76 EDITION OF 1 OCT 76 WILL BE USED UNTIL EXHAUSTED.

Source: https://assist.dla.mil - Downloaded: 2026-10-21T20:07Z
Check the source to verify that this is the current version before use.

NO.	TABLE II. (cont'd) MANUFACTURING OPERATIONS REQUIREMENTS	STITCH TYPE	SEAM AND STITCHING TYPE	STITCHES PER INCH	THREAD		
					NEEDLE	BOBBIN/ LOOPER	COVER
22.	Prepare for packing. a. Trim all ends of thread and remove loose threads. b. Remove spots and stains. c. Remove shade tickets.						

NATICK Form 903
1 Dec 76 EDITION OF 1 OCT 76 WILL BE USED UNTIL EXHAUSTED.

Source: https://assist.dla.mil -- Downloaded: 2026-10-21T20:07Z
Check the source to verify that this is the current version before use.

3.8 Measurements. The finished coverall shall conform to the measurements shown in table III.

TABLE III. Measurements (inches)

	Sizes			
	Small	Medium	Large	Tolerances
1/2 back 1/	17-1/2	18-1/2	19-1/2	± 3/4
Back elastic length 2/	15-1/2	17-1/2	19-1/2	± 1/2
1/2 waist 3/	18	20	22	± 3/4
Sleeve length 4/	25	25	25	± 1/2
Sleeve width (wrist) 5/	4-1/2	4-1/2	4-1/2	± 1/2
Leg inseam 6/	27-1/2	29-1/2	29-1/2	± 1/2
Leg width (ankle) 7/	7	7	7	± 1/2
Overall length 8/	63-1/4	65-3/8	65-3/8	± 1

1/ Measure across back, with pleats closed, from back arm seam to back arm seam.

2/ Measure elastic relaxed, from inside of coverall, from end to end.

3/ With slide fastener closed, measure across front, at waist, from folded edge to folded edge, with elastic stretched.

4/ Measure length of back arm seam.

5/ Measure through center of sleeve hem from folded edge to folded edge.

6/ Measure inseam from crotch joining seam to bottom of leg.

7/ Measure through center of leg hem from folded edge to folded edge.

8/ Measure from base of neck along center back to bottom of leg.

3.9 Workmanship. The finished coverall shall conform to the quality of product established by this specification, and the occurrence of defects shall not exceed the applicable point value or defect limit.

Source: https://assist.dla.mil -- Downloaded: 2026-10-21T20:07Z
Check the source to verify that this is the current version before use.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract or purchase order, the contractor is responsible for the performance of all inspection requirements (examinations and tests) as specified herein. Except as otherwise specified in the contract or purchase order, the contractor may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in this specification where such inspections are deemed necessary to ensure supplies and services conform to prescribed requirements.

* 4.1.1 Responsibility for compliance. All items shall meet all requirements of sections 3 and 5. The inspection set forth in this specification shall become a part of the contractor's overall inspection system or quality program. The absence of any inspection requirements in the specification shall not relieve the contractor of the responsibility of ensuring that all products or supplies submitted to the Government for acceptance comply with all requirements of the contract. Sampling inspection, as part of manufacturing operations, is an acceptable practice to ascertain conformance to requirements, however, this does not authorize submission of known defective material, either indicated or actual, nor does it commit the Government to accept defective material.

* 4.1.2 Responsibility for dimensional requirements. Unless otherwise specified in the contract or purchase order, the contractor is responsible for ensuring that all specified dimensions have been met. When dimensions cannot be examined on the end item, inspection shall be made at any point, or at all points in the manufacturing process necessary to ensure compliance with all dimensional requirements.

* 4.2 Classification of inspections. The inspection requirements specified herein are classified as follows:

a. First article inspection (see 4.3).
b. Quality conformance inspection (see 4.4).

* 4.3 First article inspection. When a first article is required (see 3.1 and 6.2), it shall be examined for defects in accordance with MIL-STD-1668.

4.4 Quality conformance inspection. Unless otherwise specified, sampling for inspection shall be performed in accordance with MIL-STD-105.

4.4.1 Component and material inspection. In accordance with 4.1, components and materials shall be inspected in accordance with all the requirements of referenced documents unless otherwise excluded, amended, modified, or qualified in this specification or applicable purchase document.

Source: https://assist.dla.mil -- Downloaded: 2026-10-21T20:07Z
Check the source to verify that this is the current version before use.

4.4.2 Point count inspection. Sampling and inspection provisions for end item examinations, dimensional examination, and packaging inspection shall be performed in accordance with MIL-STD-1668.

4.4.3 Palletization examination. The fully packaged and palletized end items shall be examined for the defects listed below. The lot size shall be expressed in units of palletized unit loads. The sample unit shall be one palletized unit load, fully packaged. The inspection level shall be S-1 and the AQL, expressed in terms of defects per hundred units, shall be 6.5.

Examine	Defect
Finished dimensions	Length, width, or height exceeds specified maximum requirement
Palletization	Pallet pattern not as specified Interlocking of loads not as specified Load not bonded as specified
Weight	Exceeds maximum load limits
Marking	Omitted; incorrect; illegible; of improper size, location, sequence, or method of application

5. PACKAGING

5.1 Preservation. Preservation shall be level A or Commercial as specified (see 6.2).

5.1.1 Level A preservation. Each pair of coveralls, with slide fasteners closed, shall be laid flat front down. Fold the leg bottoms up approximately 5 inches. Bring the leg portion up even with the shoulder line and fold the sleeves at right angles across the back. The length shall then be folded in half. Each completely folded coverall shall measure approximately 23 by 14-1/2 inches.

5.1.2 Commercial preservation. The coveralls shall be preserved in accordance with ASTM D 3951.

5.2 Packing. Packing shall be level A, B, or Commercial as specified (see 6.2).

* 5.2.1 Level A packing. Ten pairs of coveralls of one size only, preserved as specified in 5.1, shall be packed in a snug-fitting fiberboard shipping container, conforming to RSC-L, grade V2s, of PPP-B-636. The inside of each shipping container shall be fitted with a box liner conforming to type CF, class weather-resistant, variety DW, grade V15c of PPP-B-636. Level A unit packs shall be packed flat within the shipping container with every other pair reversed top to bottom and the collar facing up. Inside dimensions of each container shall be approximately 23-1/2 inches in length, 15 inches in width,

Source: https://assist.dla.mil -- Downloaded: 2026-10-21T20:07Z
Check the source to verify that this is the current version before use.

and 15 inches in depth. Approximate dimensions are furnished as a guide only. Each container shall have the contents completely covered on the top and bottom with a sheet of 30-pound minimum basis weight kraft paper conforming to A-A-203. Each shipping container shall be closed in accordance with method III, waterproofed in accordance with method V, and reinforced as specified in the appendix of PPP-B-636, except that the inspection shall be in accordance with 4.4.2. Shipping containers shall be arranged in unit loads in accordance with MIL-L-35078 for the type and class of load specified (see 6.2). When unit loads are strapped, strapping shall be limited to nonmetallic strapping, except for type II, class F loads.

* 5.2.2 Level B packing. Ten pairs of coveralls of one size only, preserved as specified in 5.1, shall be packed in a snug-fitting fiberboard shipping container, conforming to style RSC-L, type CF (variety SW) or SF, class domestic, grade 275 of PPP-B-636. The inside of each shipping container shall be fitted with a box liner conforming to type CF, class domestic, variety DW, grade 275 of PPP-B-636. Level A unit packs shall be packed flat within the shipping container with every other pair reversed top to bottom and the collar facing up. Inside dimensions of each container shall be approximately 23-1/2 inches in length, 15 inches in width, and 15 inches in depth. Approximate dimensions are furnished as a guide only. Each container shall have the contents completely covered on the top and bottom with a sheet of 30-pound minimum basis weight kraft paper conforming to A-A-203. Each shipping container shall be closed in accordance with method II as specified in the appendix of PPP-B-636, except that the inspection shall be in accordance with 4.4.2.

5.2.2.1 Weather-resistant fiberboard containers. When specified (see 6.2), the shipping container shall be a grade V3c, V3s, or V4s fiberboard box fabricated in accordance with PPP-B-636 and closed in accordance with method III as specified in the appendix of PPP-B-636, except that the inspection shall be in accordance with 4.4.2.

5.2.3 Commercial packing. Coveralls, preserved as specified in 5.1, shall be packed in accordance with ASTM D 3951.

* 5.3 Palletization. When specified (see 6.2), coveralls, packed as specified in 5.2.2 or 5.2.3, shall be palletized on a 4-way entry pallet in accordance with load type Ia of MIL-STD-147. Pallet type shall be type I (4-way entry), type IV or type V in accordance with MIL-STD-147. Pallets shall be fabricated from wood groups I, II, III, or IV of MIL-STD-731. Each prepared load shall be bonded with primary and secondary straps in accordance with bonding means C and D or film bonding means F or G. Pallet pattern shall be number 3 in accordance with the appendix of MIL-STD-147. Interlocking of loads shall be effected by reversing the pattern of each course.

Source: https://assist.dla.mil -- Downloaded: 2026-10-21T20:07Z
Check the source to verify that this is the current version before use.

5.4 Marking. In addition to any special marking required by the contract or purchase order, shipping containers and palletized unit loads shall be marked in accordance with MIL-STD-129 or ASTM D 3951, as applicable.

6. NOTES

(This section contains information of a general or explanatory nature that may be helpful, but is not mandatory.)

6.1 Intended use. The coveralls are intended to be worn by mechanics in the Department of the Army as a protective garment over the cold weather clothing.

* 6.2 Acquisition requirements. Acquisition documents must specify the following:

a. Title, number, and date of this specification.
b. Size required (see 1.2).
c. Issue of DODISS to be cited in the solicitation, and if required, the specific issue of individual documents referenced (see 2.1.1 and 2.2).
d. When a first article is required (see 3.1, 4.3, and 6.3).
e. Levels of preservation and packing (see 5.1 and 5.2).
f. Type and class of unit load required (see 5.2.1).
g. When weather-resistant grade fiberboard shipping containers are required for level B packing (see 5.2.2.1).
h. When palletization is required (see 5.3).

* 6.3 First article. When a first article is required, it shall be inspected and approved under the appropriate provisions of FAR 52.209. The first article should be a preproduction sample. The contracting officer should specify the appropriate type of first article and the number of units to be furnished. The contracting officer should also include specific instructions in acquisition documents regarding arrangements for selection, inspection, and approval of the first article.

6.4 Sample. For access to samples, address the contracting activity issuing the invitation for bids or request for proposal.

* 6.5 Subject term (key word) listing.

Clothing
Protective garment

Source: https://assist.dla.mil -- Downloaded: 2026-10-21T20:07Z
Check the source to verify that this is the current version before use.

6.6 Changes from previous issue. The margins of this specification are marked with asterisks to indicate where changes (additions, modifications, corrections, deletions) from the previous issue were made. This was done as a convenience only and the Government assumes no liability whatsoever for any inaccuracies in these notations. Bidders and contractors are cautioned to evaluate the requirements of this document based on the entire content irrespective of the marginal notations and relationship to the last previous issue.

Custodian:

Army - GL

Review activities:

Army - MD
DLA - CT

Preparing activity:

Army - GL

(Project 8415-0724)

Source: https://assist.dla.mil -- Downloaded: 2026-10-21T20:07Z
Check the source to verify that this is the current version before use.

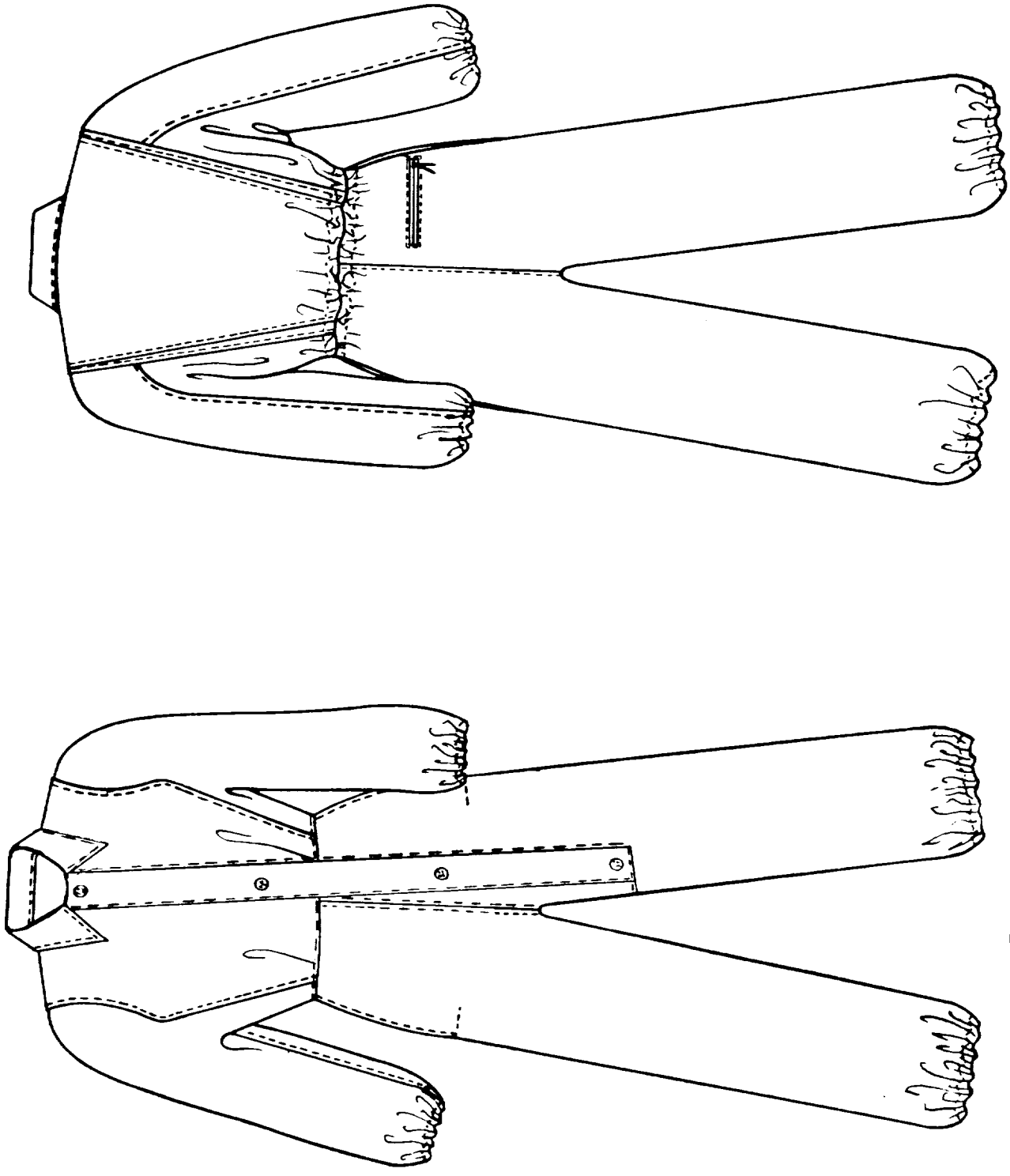


FIGURE 1. Coverall, mechanics, cold weather.

Source: https://assist.dla.mil -- Downloaded: 2026-10-21T20:07Z
Check the source to verify that this is the current version before use.

STANDARDIZATION DOCUMENT IMPROVEMENT PROPOSAL

INSTRUCTIONS

1. The preparing activity must complete blocks 1, 2, 3, and 8. In block 1, both the document number and revision letter should be given.
2. The submitter of this form must complete blocks 4, 5, 6, and 7.
3. The preparing activity must provide a reply within 30 days from receipt of the form.

NOTE: This form may not be used to request copies of documents, nor to request waivers, or clarification of requirements on current contracts. Comments submitted on this form do not constitute or imply authorization to waive any portion of the referenced document(s) or to amend contractual requirements.

I RECOMMEND A CHANGE:	1. DOCUMENT NUMBER	2. DOCUMENT DATE (YYMMDD)
	MIL-C-41833E(GL)	1990 March 30

3. DOCUMENT TITLE
Coverall, Mechanic's, Cold Weather

4. NATURE OF CHANGE *(Identify paragraph number and include proposed rewrite, if possible. Attach extra sheets as needed.)*

5. REASON FOR RECOMMENDATION

6. SUBMITTER

a. NAME (Last, First, Middle Initial)	b. ORGANIZATION	
c. ADDRESS (Include Zip Code)	d. TELEPHONE (Include Area Code) (1) Commercial (2) AUTOVON (If applicable)	7. DATE SUBMITTED (YYMMDD)

8. PREPARING ACTIVITY

a. NAME	b. TELEPHONE (Include Area Code)	
U.S. Army Natick RD&E Center	(1) Commercial 508-651-5221	(2) AUTOVON 256-5221
c. ADDRESS (Include Zip Code) Commander, U.S. Army Natick RD&E Center ATTN: STRNC-ES Natick, MA 01760-5014	IF YOU DO NOT RECEIVE A REPLY WITHIN 45 DAYS, CONTACT: Defense Quality and Standardization Office 5203 Leesburg Pike, Suite 1403, Falls Church, VA 22041-3466 Telephone (703) 756-2340 AUTOVON 289-2340	

Source: https://assist.dla.mil -- Downloaded: 2026-10-21T20:07Z
Check the source to verify that this is the current version before use.